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UPPER HEYFORD, United Kingdom

WMO: **036553**

Lat: **51.933N** Lon: **1.250W** Elev: **436** StdP: **14.47** Time Zone: **0.00 (EUW)** Period: **82-93** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	23.0	26.3	15.4	12.2	24.9	19.1	14.6	28.9	33.1	47.1	29.5	46.0	7.7	60	0.605	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	16.1	78.6	64.3	75.2	63.1	71.9	61.5	65.9	74.8	64.3	72.4	62.6	70.1	8.6	180	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	62.7	86.5	70.0	61.0	81.3	67.5	59.3	76.5	65.9	31.1	74.9	29.7	72.4	28.4	70.2	71.1	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	25.6	22.2	19.5	DB	18.6	83.9	6.6	4.9	13.8	87.4	10.0	90.2	6.3	93.0	1.5	96.5	(4)
(5)				WB	17.3	68.8	6.3	1.3	12.7	69.7	9.0	70.5	5.4	71.2	0.8	72.2	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	49.7	39.3	38.2	42.9	46.3	52.5	57.5	62.2	61.3	57.3	51.5	44.4	41.6	(6)
(7)			DBStd	9.79	6.64	6.73	4.70	4.99	5.00	5.27	4.46	4.35	4.32	5.03	5.77	6.04	(7)
(8)			HDD50	1540	332	330	225	132	27	3	0	0	2	41	183	265	(8)
(9)			HDD65	5649	797	750	685	561	388	232	112	130	233	418	618	725	(9)
(10)			CDD50	1419	0	1	4	21	104	229	378	352	221	89	15	5	(10)
(11)			CDD65	50	0	0	0	0	0	7	24	17	2	0	0	0	(11)
(12)			CDH74	403	0	0	0	0	8	46	217	115	15	2	0	0	(12)
(13)			CDH80	60	0	0	0	0	0	2	35	23	0	0	0	0	(13)
(14)	Wind		WSAvg	9.0	10.5	10.2	9.9	8.9	8.2	7.8	8.1	8.5	8.1	8.8	9.2	9.5	(14)
(15)	Precipitation		PrecAvg	27.2	2.6	1.9	1.7	2.0	2.4	1.9	2.0	2.3	2.1	2.8	2.8	2.7	(15)
(16)			PrecMax	36.5	7.1	4.4	3.5	5.6	5.6	5.3	5.6	5.7	4.4	5.0	4.9	4.7	(16)
(17)			PrecMin	17.4	0.4	0.2	0.2	0.0	0.4	0.2	0.3	0.1	0.1	0.7	1.1	0.7	(17)
(18)			PrecStd	5.0	1.3	1.1	0.9	1.4	1.1	1.3	1.2	1.4	1.3	1.2	1.1	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		53.9	55.2	59.4	66.5	75.2	78.8	84.3	82.3	75.6	67.6	59.0	55.7	(19)
(20)			MCWB		50.8	51.3	50.4	55.8	62.3	65.5	66.1	66.2	62.7	60.1	55.7	52.4	(20)
(21)		2%	DB		52.0	52.0	55.1	61.2	70.2	75.0	79.2	76.8	71.5	64.0	57.1	54.0	(21)
(22)			MCWB		49.2	48.5	49.4	52.2	59.7	63.9	64.2	63.6	61.0	58.4	54.2	50.8	(22)
(23)		5%	DB		50.3	50.0	52.0	57.7	66.1	71.4	75.5	73.5	67.9	61.1	55.2	52.2	(23)
(24)			MCWB		47.8	47.2	47.5	50.5	56.6	62.0	63.2	62.3	59.8	56.4	52.2	49.2	(24)
(25)		10%	DB		48.5	48.1	50.2	55.4	62.5	67.9	71.9	70.2	64.7	59.2	53.5	50.4	(25)
(26)			MCWB		46.4	45.9	46.7	49.2	54.6	59.3	61.6	60.8	57.8	54.8	50.6	47.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		51.8	51.7	52.5	57.4	63.8	67.3	68.9	67.7	64.6	60.9	56.1	53.5	(27)
(28)			MCDB		53.9	54.5	58.0	63.7	73.0	75.3	80.4	78.1	73.1	65.9	58.2	55.3	(28)
(29)		2%	WB		50.3	49.4	50.6	54.0	60.3	64.4	66.3	65.4	62.3	59.3	54.0	52.0	(29)
(30)			MCDB		51.9	51.4	53.3	59.8	68.2	73.0	76.7	73.6	68.5	62.9	56.3	54.5	(30)
(31)		5%	WB		48.5	47.5	48.9	51.8	57.7	62.1	64.4	63.6	61.0	57.3	52.8	50.4	(31)
(32)			MCDB		50.2	49.6	51.0	56.4	64.8	69.8	72.7	70.8	66.6	60.7	55.1	52.5	(32)
(33)		10%	WB		46.6	45.5	47.2	50.0	55.4	60.0	62.5	61.9	59.4	55.5	51.0	48.4	(33)
(34)			MCDB		48.7	47.6	49.7	54.6	61.1	66.9	70.5	68.5	64.1	58.8	53.3	50.8	(34)
(35)	Mean Daily Temperature Range		MDBR		8.5	9.3	11.3	13.7	15.2	14.8	16.1	15.6	13.7	11.1	9.1	7.9	(35)
(36)		5% DB	MCDBR		10.3	11.0	13.1	17.4	21.3	20.1	22.7	20.2	18.1	12.3	9.4	9.0	(36)
(37)			MCWBR		10.4	9.7	9.6	11.2	12.9	11.5	11.1	10.0	10.4	8.8	8.5	8.7	(37)
(38)		5% WB	MCDBR		10.5	10.3	11.7	16.2	19.8	19.0	19.6	18.3	15.8	11.4	9.8	8.9	(38)
(39)			MCWBR		11.0	9.4	9.7	11.3	12.7	11.5	10.9	10.2	10.6	9.2	9.2	9.0	(39)
(40)	Clear-Sky Solar Irradiance		taub		0.320	0.336	0.369	0.393	0.398	0.396	0.393	0.382	0.365	0.349	0.323	0.310	(40)
(41)			taud		2.368	2.385	2.287	2.233	2.255	2.285	2.301	2.355	2.402	2.433	2.414	2.349	(41)
(42)			Ebn at Noon		216	247	259	266	271	272	271	268	259	239	213	197	(42)
(43)			Edh at Noon		20	25	33	40	41	40	39	35	30	24	18	17	(43)
(44)	All-Sky Solar Radiation		RadAvg		251	464	804	1236	1510	1647	1571	1301	979	570	303	200	(44)
(45)			RadStd		21	59	105	141	96	166	162	101	86	50	34	23	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (95 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page